SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_MN_Duluth.Intl.AP-Duluth.ANGB.727450_TMYx.epw
- Building Name: SmallOffice
- Building Location: Duluth, MN
- Total Floor Area: 511 m^2

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof	**Wall upgrade:** • Exterior wall insulation: Expanded Polystyrene (EPS) Foam Board, target R-13.0 (skipped: target R-value already satisfied) **Roof upgrade:**

		• Roof insulation: Extruded Polystyrene (XPS) Foam Board, target R-24.4
Scenario 2	wall + roof	**Wall upgrade:** • Exterior wall insulation: Fiberglass Batts, target R-20.0 **Roof upgrade:** • Roof insulation: Blown Fiberglass, target R-30.0
Scenario 3	roof	**Roof upgrade:** • Roof insulation: Blown Fiberglass, target R-30.0

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	236.54	234.51	**2.03**	**0.86%**
Scenario 2	Total Site Energy (GJ)	236.54	233.46	**3.08**	**1.30%**
Scenario 3	Total Site Energy (GJ)	236.54	234.42	**2.12**	**0.90%**

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison

Baseline $9,841

Scenario 2 $9,703

Baseline Scenario 2

Max saving: **$-138 (-1.40%)**

Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr) Comparison

Baseline 20,894

Scenario 2 20,660

Baseline Scenario 2

Max saving: **-234 kg CO_2e (-1.12%)**

Min Retrofit Construction Cost

$7,922

Scenario 3

Min Retrofit Embodied Carbon

8,391 kg CO_2e

Embodied carbon intensity (ECI): 16 kgCO_2e/m^2

Scenario 3

Lowest Retrofit Cost Payback Period

61 years

Scenario 2

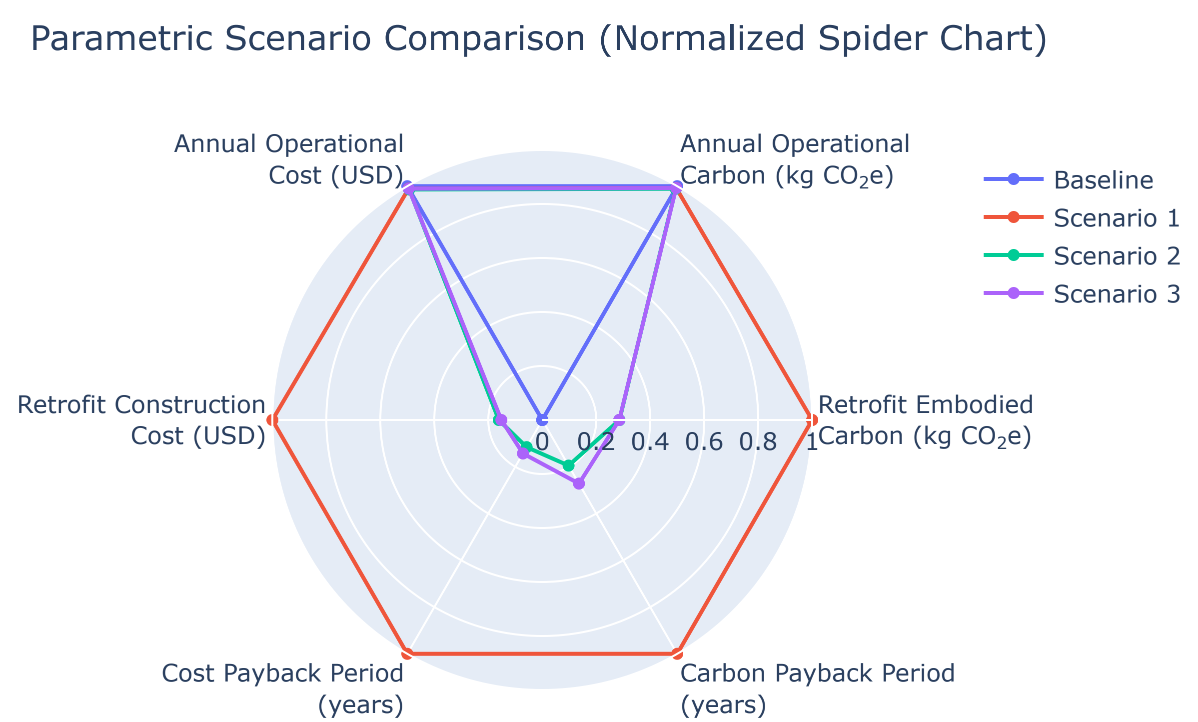
Lowest Retrofit Carbon Payback Period

36 years

Scenario 2

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

Scenario 1 526 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr)).

Scenario 1 185 yrs

Scenario 2 61 yrs

Scenario 3 75 yrs

Scenario 2 36 yrs

Scenario 3 50 yrs

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

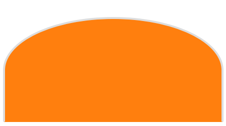
Scenario	Component	Material name	Volume (m^3)	Area (m^2)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m^3)	Product Lifetime (years)
Scenario 1	Wall insulation	Expanded Polystyrene (EPS) Foam Board	N/A	N/A	N/A	N/A	0.03	20	60
Scenario 1	Roof insulation	Extruded Polystyrene (XPS) Foam Board	101	599	0.17	N/A	0.03	26	75
Scenario 2	Wall insulation	Fiberglass Batts	4.21	282	N/A	N/A	0.03	30	60
Scenario 2	Roof insulation	Blown Fiberglass	125	599	0.21	N/A	0.03	30	68
Scenario 3	Roof insulation	Blown Fiberglass	125	599	0.21	N/A	0.03	30	68

Result Summary

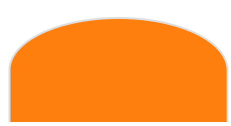
Per-scenario retrofit contribution breakdown. Each scenario includes two pie charts: cost and embodied carbon by retrofit measure.

Scenario 1

Cost breakdown by retrofit measure



Carbon breakdown by retrofit measure



Roof: $52,261 (100.0%)

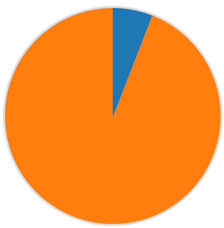
Total: $52,261

Roof: 29,412 (100.0%)

Total: 29,412 kg CO_2e

Scenario 2

Cost breakdown by retrofit measure

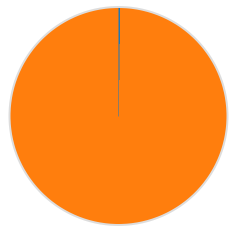


Wall: $499 (5.9%)

Roof: $7,922 (94.1%)

Total: $8,420

Carbon breakdown by retrofit measure



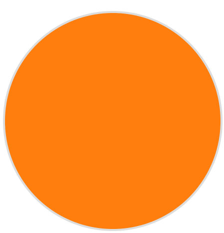
Wall: 19 (0.2%)

Roof: 8,391 (99.8%)

Total: 8,410 kg CO_2e

Scenario 3

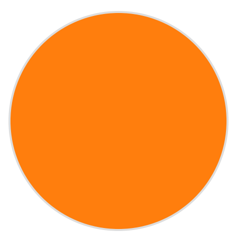
Cost breakdown by retrofit measure



Roof: $7,922 (100.0%)

Total: $7,922

Carbon breakdown by retrofit measure



Roof: 8,391 (100.0%)

Total: 8,391 kg CO_2e

Scenario	Retrofit Embodied Carbon (kg CO_2e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO_2e)	Annual Operational Cost (USD)
Baseline	0.00	$0.00	20,894	$9,841

Scenario 1	29,412	$52,261	20,735	$9,742
Scenario 2	8,410	$8,420	20,660	$9,703
Scenario 3	8,391	$7,922	20,727	$9,736

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